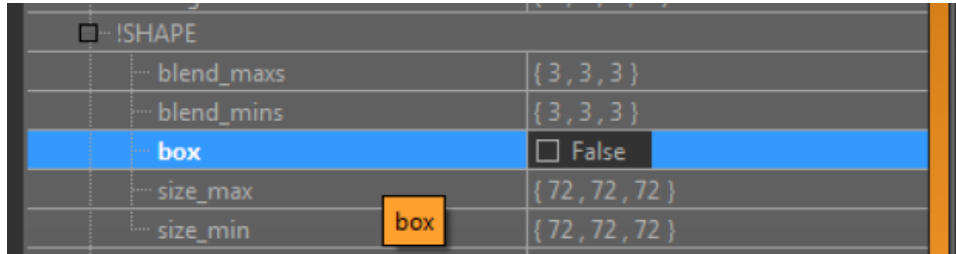


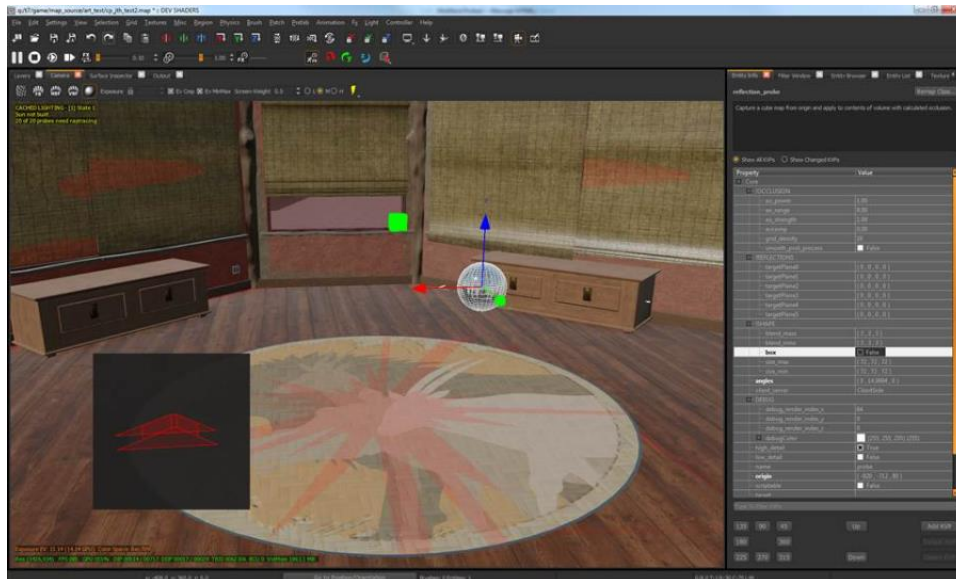
Multiface Probes

This feature allows you to toggle a parent probe or multi-box child from "box" shape to multi-face. Each multi-face probe can have up to 24 faces, and multi-face probes can be used as multi-box children allowing more complex shapes to be created. Under the hood, in the shader, the probe is still split up into multiples of 6 faces, so a full 24-face probe is roughly equivalent in performance to a box-shaped parent with 3 box-shaped children. Probes can be toggled into this mode by turning off the "box" checkbox in entity properties:



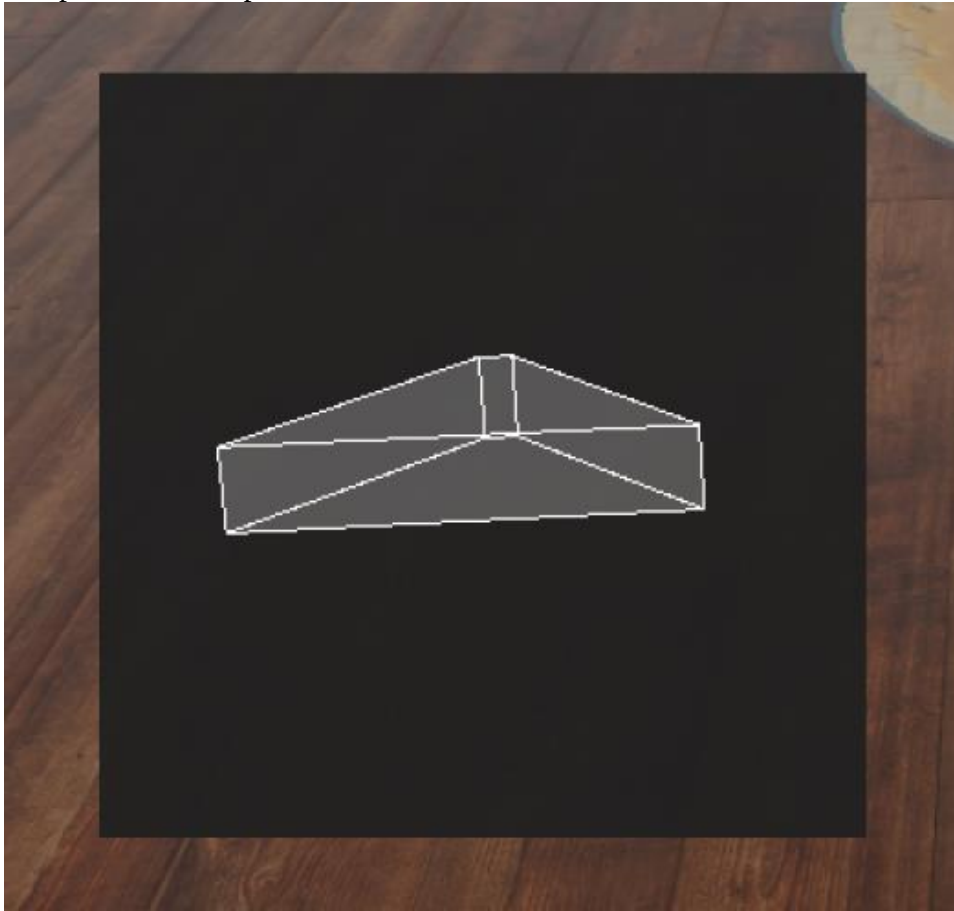
Creating Faces

Each multi-face probe can only be a convex shape, the same way regular brushes work. That means they are defined by a collection of planes. When you first switch a probe over to multi-face mode, it has no planes, and therefore no shape. Planes can be added to the shape by Alt-Clicking on surfaces (the same binding as the auto-snap resize, called "Attempt Soft Selection" in the properties menu). Alt-Clicking again on an existing face will remove it. This only works if you have only one object selected. It's important to remember to add faces by clicking on the surface that point in toward the inside of the probe. Clicking on a surface that point away from the probe will add a face that hides all the other faces. As faces are added to the shape, the overall shape is shown in a little picture-in-picture in the corner of the 3D window.

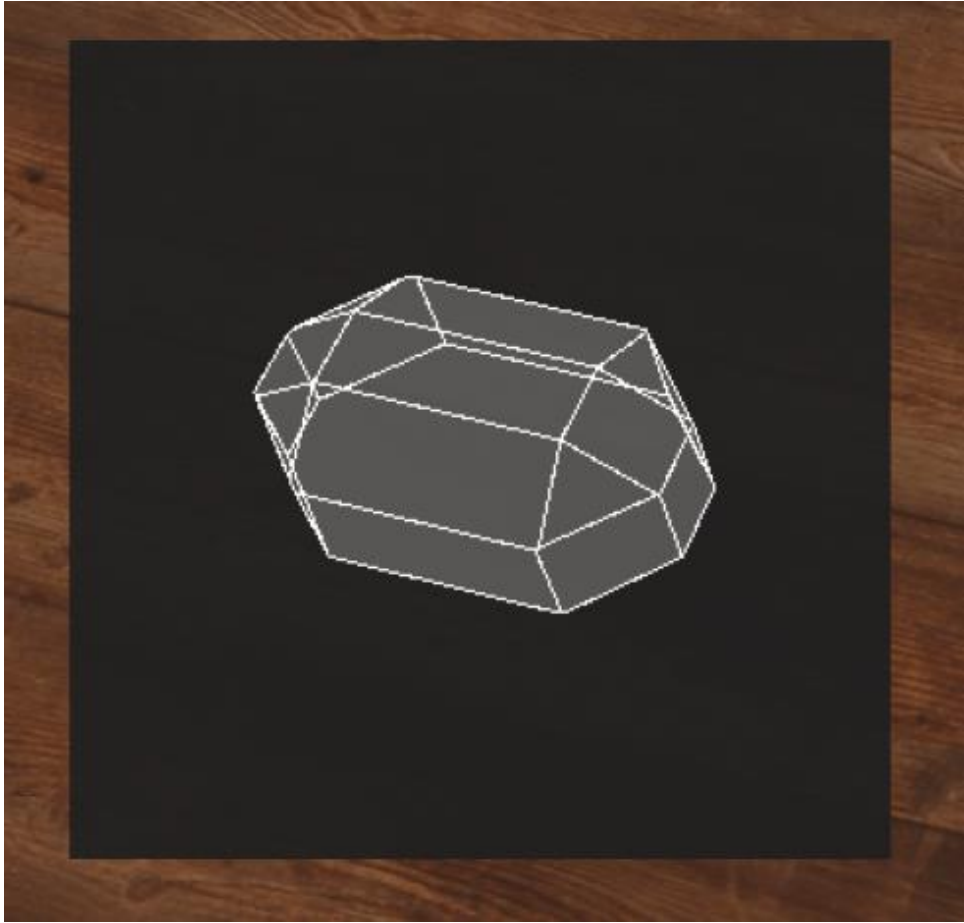


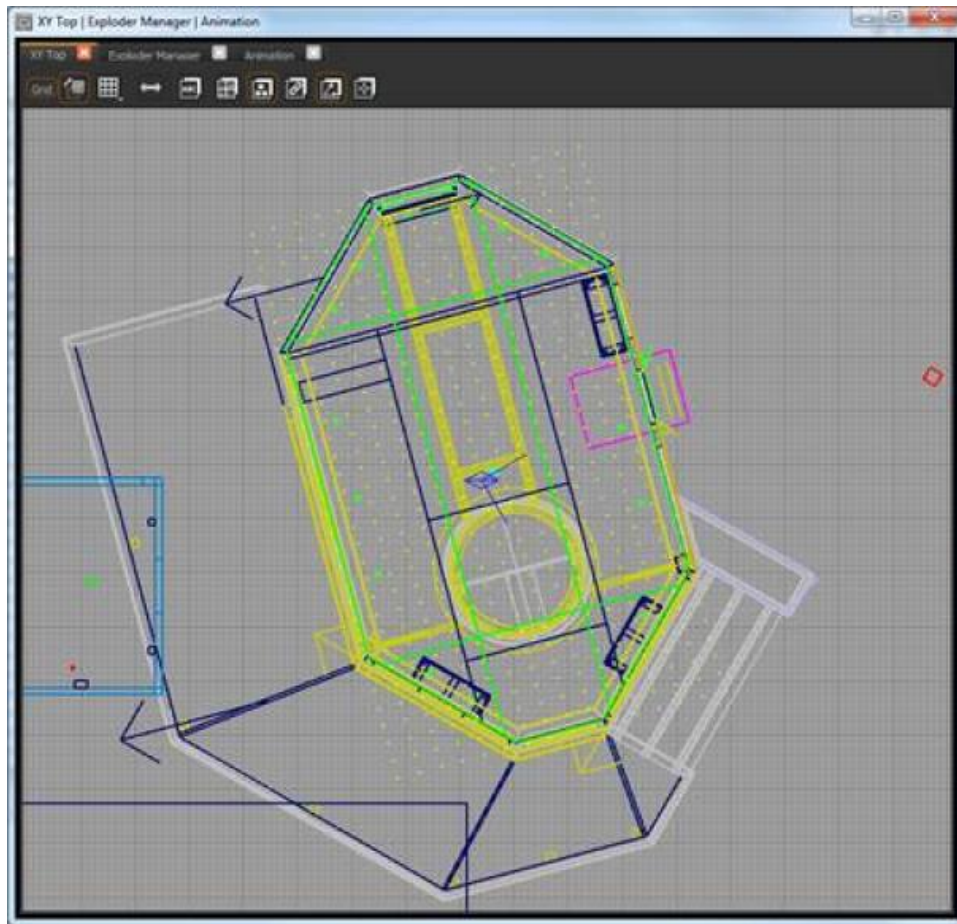
Once enough planes are added to close the shape off, the mini-map display will change from red

to white, and the probe face colors will go back to normal to show you're dealing with a complete valid shape.



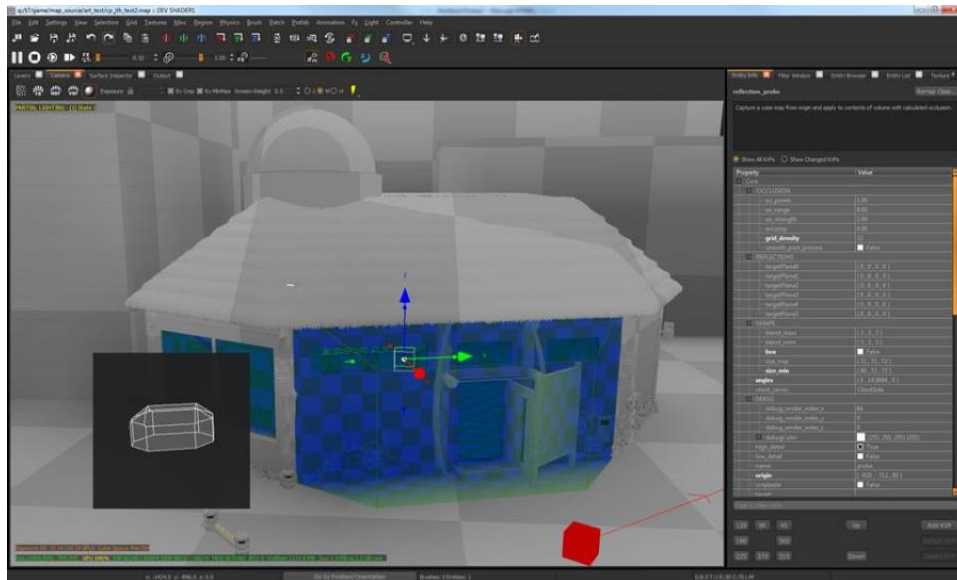
Below is the completed shape for the room shown above. It went from two probes with 7 multi-boxes each, to a single probe with 18 faces, which is the equivalent of one probe with two multi-boxes. The mini-map helps you ensure you've created all the faces you need. One important note is to make sure your probe is aligned to the room before creating the faces. This will ensure the tightest fit with the volume texels, and the best lighting quality.





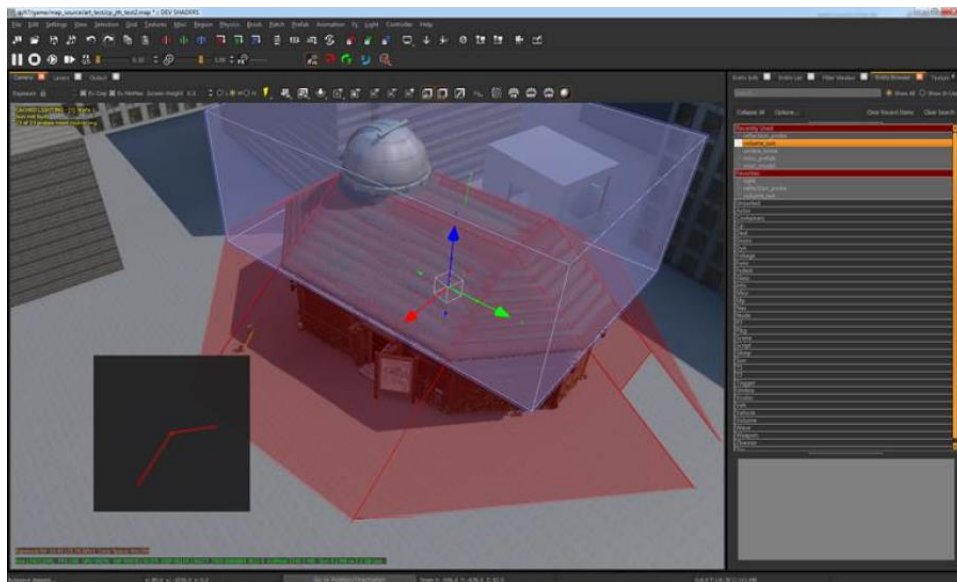
Editing Faces and Blends

Existing faces can be edited by Ctrl+Dragging them in and out in the 3D window (the "Modify Face" action in preferences). If you use Ctrl+Click to select objects, you'll need to deselect your probe before selecting other objects. Each face has a blend, the same way box probes do. The blends can be modified by Ctrl+Alt+Right+dragging them (the "Rotate Face Texture" action in preferences).



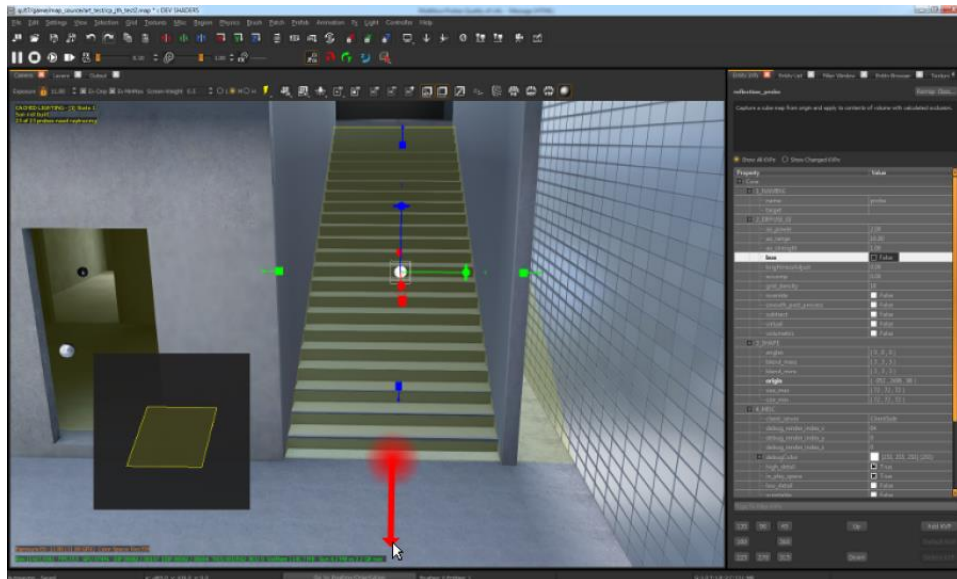
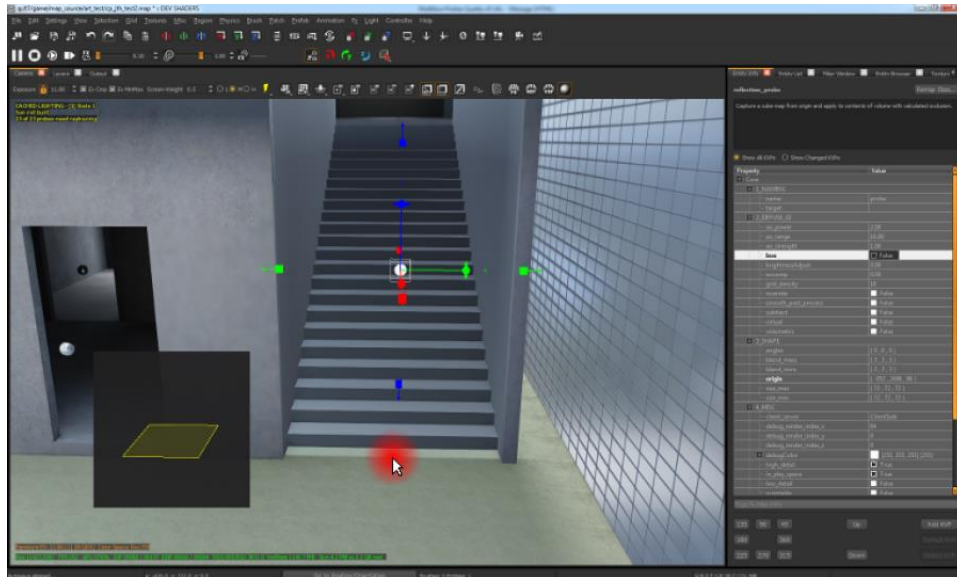
Multiface Probes Quality of Life

Snapping and face creation against unselected probes, when "Show Unselected Probe Boxes" is on and works against the faces of unselected multiface probes. There robust geometry creation for un-closed shapes, guaranteeing that all planes that contribute to the shape get at least some visible face in the mesh that's drawn in viewport. This is especially important for subtract probes, since they're usually left un-closed.



If you hit "R" to go into rotate mode, then Ctrl+Drag a face on a multiface probe, the face will pivot around the point you dragged from. This way you can rotate existing faces into position

instead of making placeholder brushes to snap to. This works with grid snap, making it easier to snap to exact angles along only one axis and to rotate multiple times around the same point.



When Alt+Dragging to create a new face for a multiface probe in the 2D window, a preview line is drawn from the mouse making it easier to line up the face, especially when your first click doesn't result in visible geometry. The solid thick line is the line you're drawing with your mouse, and the dashed line is extended from both sides. The new face will be created along this line.

